From Engineering Mathematics to Rigorous Pure Mathematics

A Self-Study Curriculum for a Software Engineer

Prepared for Zim

May 2026

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1 How to Use This Curriculum

The first goal is not speed. The first goal is to become the kind of reader who can sit with a definition, test it against examples, see why each word is there, and then write a proof that another careful person can check.

The initial anchor text is Daniel Velleman's *How to Prove It*. This curriculum prepares you to read it with less fog, then uses it as the first serious proof text. The material here is original and is meant to support the book, not replace it.

Focus

Each week has five parts: read, translate notation, prove small claims, debug one proof, and review. If a week feels too fast, split it into two. Mathematical maturity is built by returning to the same ideas with sharper eyes.

1.1 The Working Rhythm

1. Read slowly with pencil in hand. Rewrite every definition in your own words.
2. Build a tiny example and a tiny non-example.
3. Try the proof before reading a polished solution.
4. Mark every proof gap with a concrete label: unknown definition, unjustified implication, hidden case, unclear quantifier, or algebra slip.
5. Review old prompts on days 1, 3, 7, 14, and 30.

2 The Map of Mathematics

Engineering mathematics often asks, "What method solves this problem?" Pure mathematics asks, "What structure is present, and what follows from the assumptions?"

Area	Central objects	Why it matters later
Logic	statements, quantifiers, implication	proof grammar for every later field
Sets and functions	membership, maps, images, inverse images	language of analysis, algebra, topology, geometry
Relations	equivalence, order, partitions	quotient spaces, congruences, bases, classification
Number systems	$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	examples where proof methods become visible
Real analysis	limits, continuity, completeness	makes calculus honest and trains epsilon thinking
Abstract algebra	groups, rings, fields, modules	symmetry, equations, coding theory, cryptography, geometry
Topology	open sets, continuity, compactness	the portable skeleton of analysis and geometry
Differential geometry	manifolds, tangent spaces, curvature	calculus on curved spaces, physics, optimization

Algebraic geometry	varieties, ideals, schemes	shape described by polynomial equations

2.1 Prerequisite Dependency Graphs

```
Logic -> Sets -> Functions -> Relations -> Proof fluency
Proof fluency -> Induction -> Countability -> Construction of number systems
Sets + Functions + Inequalities -> Sequences -> Limits -> Continuity
Limits + Completeness -> Real Analysis -> Metric Spaces -> Topology
Relations + Functions -> Groups -> Rings -> Fields -> Algebraic Geometry
Linear Algebra + Analysis -> Multivariable Analysis -> Differential Geometry
Topology + Commutative Algebra -> Algebraic Geometry
```

3 Notation and Logic Bootcamp

Definition 1. A *statement* is a sentence that is either true or false. A *predicate* is a sentence with variables that becomes a statement after the variables are assigned values.

Intuition

Think of a predicate like a function returning a Boolean. In code, `isEven(n)` is not true or false until `n` is specified. In mathematics, $P(n)$ works the same way, except the domain and assumptions must be explicit.

Formal

The statement $\forall n \in \mathbb{Z},\ n^2 \geq 0$ says every integer has a nonnegative square. The statement $\exists n \in \mathbb{Z},\ n^2 = 2$ says at least one integer squares to 2, which is false.

3.1 Core Symbols

Symbol	Meaning	Read it as
$\forall$	universal quantifier	for every
$\exists$	existential quantifier	there exists
$\in$	membership	is an element of
$\subseteq$	subset	every element of the left set is in the right set
$\Rightarrow$	implication	if left, then right
$\Leftrightarrow$	biconditional	if and only if
$\neg P$	negation	not P
$P \wedge Q$	conjunction	P and Q
$P \vee Q$	disjunction	P or Q, inclusive
$f : A \to B$	function	f maps each element of A to an element of B
$f^{-1}(S)$	inverse image	all inputs whose outputs land in S

3.2 Quantifier Negation

Theorem 1. *For any predicate $P(x)$ on a domain D,*

$$\neg(\forall x \in D,\ P(x)) \iff \exists x \in D,\ \neg P(x).$$

Also,

$$\neg(\exists x \in D,\ P(x)) \iff \forall x \in D,\ \neg P(x).$$

Intuition

To disprove "every bridge in town is open," you need one closed bridge. To disprove "some bridge is open," every bridge must be closed.

4 Weekly Study Plan: Bridge Into Velleman

Week	Theme	Main work	Exercises	Milestone
1	Statements and notation	Learn propositions, predicates, truth tables, domains	Translate 30 English statements into symbols and back	You can tell statement from predicate
2	Quantifiers	Master $\forall, \exists$, nested quantifiers, negation	Write negations without the phrase "it is not true that"	Quantifier fog begins to lift
3	Direct proof	Prove implications by assuming the hypothesis	Divisibility, parity, inequalities	First clean one-page proof
4	Contrapositive and contradiction	Learn when each method reduces friction	Irrationality of $\sqrt{2}$, parity converses	You can choose a proof strategy
5	Sets	Subsets, union, intersection, complement, power set	Prove set identities element by element	Set notation feels like grammar
6	Functions	Domain, codomain, injective, surjective, bijective	Prove or refute claims about images and inverse images	You stop treating f^{-1} as only an inverse function
7	Relations	Equivalence relations, orders, partitions	Modular congruence, divisibility order	Quotients become less mysterious
8	Induction	Weak, strong, structural induction	Sums, inequalities, recursive sequences	You can write induction without hand waving
9	Velleman launch	Begin *How to Prove It*, Chapters 1-2	Redo early exercises in full sentences	Book reading becomes active
10	Velleman logic	Conditional proof, biconditionals, cases	Build proof outlines before final proofs	You can see proof architecture
11	Velleman sets	Set proofs, indexed families, power sets	Prove De Morgan laws for sets	Comfortable with element chasing
12	Velleman functions	Composition, inverses, image/preimage	Prove composition preserves injectivity	Function proofs become routine

13	Velleman relations	Equivalence classes and partial orders	Prove congruence mod n is an equivalence relation	Quotient language opens
14	Velleman induction	Strong induction and recursively defined objects	Fibonacci divisibility, binary strings	You can prove recursive facts
15	Cardinality	Countable sets, diagonal arguments	Show $\mathbb{Q}$ is countable, $\mathbb{R}$ is uncountable	Infinity becomes structured
16	Review project	Write a mini portfolio of 12 proofs	Revise old proofs with debugger rubric	Ready for analysis/algebra entry texts

5 Weekly Study Plan: After Velleman

Week	Track	Study target	Milestone
17	Real analysis	Ordered fields, supremum property, absolute value	Prove basic inequalities from definitions
18	Real analysis	Sequences and convergence	Write epsilon proofs for simple sequences
19	Real analysis	Limits and continuity	Prove continuity using ε-δ
20	Real analysis	Compactness on $\mathbb{R}$	Understand why closed and bounded matters
21	Algebra	Groups, examples, subgroups	Prove a subset is or is not a subgroup
22	Algebra	Homomorphisms and kernels	See kernels as algebraic fibers
23	Algebra	Cosets and quotient groups	Prove Lagrange's theorem carefully
24	Algebra	Rings, ideals, polynomial rings	Connect ideals to congruence
25	Topology	Metric spaces and open balls	Rebuild continuity without formulas
26	Topology	Open sets, closed sets, bases	Prove equivalent definitions of continuity
27	Topology	Compactness and connectedness	Translate interval facts into topology
28	Topology	Quotient spaces and product spaces	Recognize where algebra and topology rhyme
29	Differential geometry	Curves, surfaces, charts	See a chart as a local coordinate API
30	Differential geometry	Tangent vectors and differentials	Compute tangent maps and interpret them
31	Differential geometry	Forms, integration, curvature preview	Connect local calculation to global shape
32	Algebraic geometry	Affine varieties and polynomial ideals	Translate equations into geometric sets
33	Algebraic geometry	Coordinate rings and maps	See functions on shapes as algebra
34	Algebraic geometry	Hilbert basis theorem preview	Understand why finite generation is powerful

35	Synthesis	Write a map of the fields	Explain how one theorem appears in three languages
36	Portfolio	Ten polished proofs across fields	Ready for serious first courses

6 Theorem Summaries to Carry Forward

Theorem	Statement	What to notice
Contrapositive	$P \Rightarrow Q$ is equivalent to $\neg Q \Rightarrow \neg P$	Often turns a vague goal into concrete information
Euclid's Lemma	If p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$	The seed of unique factorization
Well-ordering of $\mathbb{N}$	Every nonempty subset of $\mathbb{N}$ has a least element	The engine behind induction
Pigeonhole principle	More objects than boxes forces a collision	Counting can prove existence without construction
Cantor's theorem	No set has the same cardinality as its power set	Infinity has levels
Intermediate value theorem	Continuous functions on intervals hit intermediate values	Topology hiding inside calculus
Heine-Borel	In $\mathbb{R}^n$, compact means closed and bounded	A local-to-global machine for analysis
Lagrange's theorem	For finite group G, $\lvert H\rvert$ divides $\lvert G\rvert$ for every subgroup H	Symmetry has arithmetic constraints
First isomorphism theorem	$G/\ker f \cong \operatorname{im} f$	Quotient by what a map cannot see
Basis theorem preview	Polynomial ideals over a field are finitely generated	Algebraic geometry becomes finite enough to study

7 Formal Definitions With Friendly Pictures

7.1 Function

Formal

A function $f : A \to B$ assigns to each $a \in A$ exactly one element $f(a) \in B$.

Intuition

The domain is not decoration. It is part of the machine. The formula x^2 behaves differently as a function $\mathbb{R} \to \mathbb{R}$, $\mathbb{N} \to \mathbb{N}$, or $\mathbb{C} \to \mathbb{C}$.

7.2 Injective and Surjective

Formal

$f : A \to B$ is injective if $\forall a_1, a_2 \in A,\ f(a_1) = f(a_2) \Rightarrow a_1 = a_2$. It is surjective if $\forall b \in B, \exists a \in A$ such that $f(a) = b$.

Intuition

Injective means the function does not merge distinct inputs. Surjective means every target point is actually hit. In software terms, injectivity protects identity; surjectivity guarantees coverage of the declared output space.

7.3 Open Set

Formal

In a metric space (X, d), a set $U \subseteq X$ is open if for every $x \in U$, there exists $r > 0$ such that $B_r(x) \subseteq U$.

Intuition

Every point of an open set has wiggle room. You can move a little without leaving the set. That small phrase becomes the foundation for continuity, manifolds, and topology.

8 Proof-Writing Drills

8.1 Daily Ten-Minute Drill

1. Write the claim in symbols.
2. Circle the quantifiers.
3. Identify the hypothesis and conclusion.
4. List definitions that appear in the claim.
5. Decide the proof strategy: direct, contrapositive, contradiction, cases, induction, or construction.
6. Write a three-line proof skeleton.
7. Fill only one gap.
8. Read the proof aloud and remove any sentence that says “clearly” without evidence.

8.2 Walking Questions

Walking question

If a theorem says "for every," what would a counterexample look like? Can you picture the single object that would break it?

Walking question

When a definition has three clauses, which clause is easiest to forget? What example satisfies two clauses but fails the third?

Walking question

What is the difference between proving $A \subseteq B$ and proving $A = B$? Where do the two directions live?

Walking question

What does a function remember, and what does it forget? In algebra, that question becomes the idea of a kernel.

9 Proof Debugger Workflow

When you submit or review a proof, use this structure.

1. **Claim parsed:** Rewrite the theorem with all quantifiers explicit.
2. **Definitions needed:** List every definition used.
3. **Strategy:** Name the method and why it fits.
4. **Line check:** For each line, ask what justifies it.
5. **Gap labels:** Mark any issue as quantifier gap, definition gap, implication gap, case gap, example gap, or notation gap.
6. **Minimal repair:** Fix the smallest broken step before rewriting the whole proof.
7. **Final polish:** Remove computational clutter and keep the logical spine visible.

Focus

The best feedback is specific. "This proof is wrong" is smoke. "Line 4 assumes the conclusion for an arbitrary x, but x was chosen from a smaller set" is fire you can put out.

10 Milestones Toward Higher Mathematics

10.1 Real Analysis

You are ready when you can negate nested quantifiers, prove inequalities from definitions, and write epsilon proofs without guessing the epsilon at the end. Analysis is where the infinite becomes accountable.

10.2 Abstract Algebra

You are ready when functions, equivalence relations, and modular arithmetic feel natural. Algebra studies operations and symmetry. A group is not a bag of elements; it is a world where composition behaves predictably.

10.3 Topology

You are ready when open sets, inverse images, and proof by definition feel comfortable. Topology studies what survives stretching, bending, and continuous deformation. It is the part of analysis that stopped depending on distance.

10.4 Differential Geometry

You are ready when linear algebra, multivariable calculus, and topology have become friends. A manifold is a space that may be globally curved or complicated but locally looks like $\mathbb{R}^n$.

10.5 Algebraic Geometry

You are ready when rings, ideals, quotient structures, and topology no longer sound like separate countries. Algebraic geometry studies geometric shapes by the polynomial equations that define them.

11 Historical Thread

Proof culture did not appear all at once. Euclid organized geometry into axioms and deductions. Fermat, Euler, Gauss, Cauchy, Weierstrass, Noether, Hilbert, Grothendieck, and many others changed what mathematicians considered a satisfying explanation. Calculus began as a powerful engine for motion and change; analysis later rebuilt its foundations. Algebra began with solving equations; it became the study of structure. Geometry began with measurement; it became the study of spaces, transformations, and local coordinates.

This matters because the fields are not arbitrary school subjects. They are repairs and expansions. When a method works too well to ignore but too vaguely to trust, mathematics often responds by inventing a sharper language.

12 What to Focus On

- Quantifiers are the steering wheel.

- Definitions are not vocabulary; they are tools with exact operating instructions.
- Examples prevent you from floating away.
- Counterexamples teach the boundary of a theorem.
- A proof is a chain of justified transitions, not a performance of confidence.
- The same ideas recur: structure, maps, invariants, quotients, limits, compactness, symmetry.